Updated Lamport-style Proof Supplement

Metropolis Random Walks Escape Deep Basins
Faster in Higher Dimensions

Companion to the latest supplied LaTeX manuscript

Corrected manuscript and supplement, 10 November 2026

Purpose. This supplement renders the arguments of the corrected manuscript as hierarchical claims with explicit justifications and dependencies. It accompanies the 10 November 2026 source revision and supersedes the earlier supplements for that source.

Revision summary. The manuscript now includes the missing normalized-sum limit proving Remark 3, repairs its malformed multiplication command, clarifies the unbiased endpoint, adds the notebook/supplement repository link, and improves the AI-use acknowledgment. The paper's main proof route is unchanged: quotient calculation, shell-shorting lower bound, capacity identity, nested-cube transport, and pyramid-flow upper bound.

Verification status. The former GAP-001 is closed by the geometric-tail argument explicitly added to the source. The exact external countable-state and infinite-network background remains conditional under this review's source boundary, as do the ancillary external comparisons. No counterexample or invalid central counting, transport, conservation, or energy calculation was identified.

Method and confidence. The conversion, forward audit, and reverse audit follow the WWresearch/lamport-proof v0.2.0 workflow. These are instructions for structured mathematical review, not a theorem checker. One assistant performed the passes sequentially; no independent reviewer or proof-assistant kernel checked the result. The companion exact-rational script passes 176 finite diagnostic cases.

Coverage. All four theorems, both propositions, both lemmas, shell counting and proposal averages, low-dimensional substitutions, the unbiased quotient identity, Remark 3, and the symmetry basis of the numerical method. Numerical datasets, empirical error claims, and literature summaries are not reverified here. Commented-out source is excluded.

Contents

I Source-mapped structured proofs **3**

1 Rendering status **3**

2 Frozen source contract **3**

3 Source inventory **3**

4 Lamport-style rendering **4**
4.1 Geometry and shell probabilities . . . 4
4.2 Theorem 1: exact quotient escape time . . . 5
4.3 Special cases and Remark 2 . . . 6
4.4 Proposition 1: reversible shell quotient . . . 6
4.5 Theorem 2: quotient asymptotic order . . . 7
4.6 Remark 3: sharper leading constant . . . 8
4.7 Lemma 1: hitting time via capacity . . . 8
4.8 Theorem 3: true-chain lower bound . . . 10
4.9 Lemma 2: transport on nested cubes . . . 11
4.10 Proposition 2: resistance upper bound . . . 12
4.11 Theorem 4: true-chain asymptotic scaling . . . 13
4.12 Ancillary symmetry and numerical-method identities . . . 14

5 Source-to-step mapping ledger **15**

6 Gap and external-support register **17**

7 Frozen audit handoff **18**

II Forward audit **18**

8 Verdict and theorem contracts **18**

9 Proof structure and findings **18**

10 Step ledger **19**

11 Exact-arithmetic diagnostics **20**

12 Minimal clarification plan **20**

III Reverse dependency audit **21**

13 Audit boundary **21**

14 Obligation graph **21**

15 Critical findings **21**

16 Dependency table **21**

17 Verdict **23**

A Repository use and provenance **23**

Part I

Source-mapped structured proofs

1 Rendering status

SOURCE-MAPPED for the declared mathematical scope. This status concerns source mapping only; it is not a validity verdict. Local proof steps are normalized with the new leading-constant proof explicitly mapped to the corrected source. Unavailable external theorem statements remain marked.

2 Frozen source contract

The authoritative input is the corrected `source/paper.tex` included with the repository package, derived from the supplied `paper (2).tex` with the authorized revisions. Its SHA-256 is

```
259bf025e882a143fe3858bb88865bb0
5ffca4c7616d186f84800407e4b55c59
```

D, N are positive integers. Exact quotient formulas allow $0 < p \leq 1$, with $\alpha = 1/p$; deep-basin and resistance results assume fixed D and fixed $0 < p < 1$. States lie in $\mathbb{Z}^D$, energy is $V(x) = \|x\|_\infty$, and proposals are uniform over $2D$ nearest neighbors. An uphill proposal is accepted with probability p; other moves are accepted. Time counts attempted moves, including rejections. Escape is first arrival anywhere on $A_N = \{x : \|x\|_\infty = N\}$.

$Q_D(N)$ denotes the mean for the separately defined shell-quotient chain; $H_D(N) = \mathbb{E}_0\tau_N$ denotes the true mean. They agree in dimension one; no general inequality $Q_D(N) \leq H_D(N)$ is assumed or proved here. Quotient resistance is a lower bound on true resistance, which is a different assertion.

Use the unnormalized measure $\mu(x) = p^{\|x\|_\infty}$ and conductance $c(x, y) = \mu(x)P(x, y)$. Dirichlet energy is $\frac{1}{2}\sum_{x\sim y} c(x, y)(f(x) - f(y))^2$, counting both orientations. Flow energy counts each unoriented edge once; divergence is outgoing signed current. Self-loops affect holding times but have zero energy. Capacity is the infimum of energy over $f(a) = 1$, $f|_A = 0$. The transport lemma's u_k denotes a distribution; the shell calculation's u_k denotes a scalar proposal probability.

Background boundary. Finite counting, algebra, elementary positive-series/ratio arguments and explicitly stated finite-chain first-step identities are checked locally. The exact texts behind the invoked countable-state recurrence theorem, infinite-network finite-target duality, Kemeny–Snell aggregation criterion, and predecessor parameter comparison were not supplied or newly verified. They remain EXT-001 through EXT-004, rather than assumed verified because they are familiar.

3 Source inventory

Source locations use stable section names and LaTeX labels. Numbers (2)–(31) follow the paper's existing equation sequence; labels are authoritative if recompilation changes numbering. Each unit contract and each numbered step has a corresponding entry in the machine-readable ledger.

Unit	Source segment and role	Placement / source label
G	III-A, III-C; Eqs. (2)–(9); Geometry and shell probabilities	eq:MDs, eq:SD, eq:uk, eq:vk
T1	IV-A–B; Eqs. (10)–(17); Theorem 1: exact quotient escape time	thm:exact-escape, eq:T0-exact
S	IV-C; Eqs. (18)–(20), Remark 2; Special cases and Remark 2	eq:T0-1D, eq:T0-2D, eq:T0-3D, rmk:unbiased

Unit	Source segment and role	Placement / source label
P1	V-A; Eqs. (21)–(22); Proposition 1: reversible shell quotient	prop:quotient, eq:conductance
T2	V-B; Eqs. (23)–(24); Theorem 2: quotient asymptotic order	thm:asymptotic, eq:theta-scaling
R3	V-B; Eq. (25); Remark 3: sharper leading constant	rmk:sharp, eq:sharp-asymptotic
L1	V-C; Eqs. (26)–(27); Lemma 1: hitting time via capacity	lem:hitting, eq:hitting-identity, eq:hitting-bounds
T3	V-D; Eq. (28); Theorem 3: true-chain lower bound	thm:lower-bound, eq:lower-bound
L2	V-E; Eq. (29); Lemma 2: transport on nested cubes	lem:transport, eq:transport-energy
P2	V-E; Eq. (30); Proposition 2: resistance upper bound	prop:resistance-upper, eq:Reff-upper
T4	V-E; Eq. (31); Theorem 4: true-chain asymptotic scaling	thm:true-scaling, eq:true-scaling
N	II; III-C; IV-C Remark 1; VI-A; Ancillary symmetry and numerical-method identities	II aggregation; III-C; rmk:1D; VI-A

4 Lamport-style rendering

4.1 Geometry and shell probabilities

ASSUME / PROVE. For integers $D \geq 1$, $k \geq 1$, the shell cardinality and the uniform shell-averaged proposal probabilities are $S_D(k) = (2k+1)^D - (2k-1)^D$, $u_k = (2k+1)^{D-1}/S_D(k)$, and $v_k = (2k-1)^{D-1}/S_D(k)$. Set $S_D(0) = 1$ separately.

Source: III-A, III-C; Eqs. (2)–(9). **Forward:** PASS. **Reverse:** FOLLOWS.

G.1. The number of points with exactly s boundary coordinates is $M_{D,s}(k) = \binom{D}{s} 2^s (2k-1)^{D-s}$, $1 \leq s \leq D$.

Justification. Choose the s coordinates, their signs, and then the interior values of the remaining coordinates. These choices identify each point exactly once.

Source: III-A; (2); SRC-001.

G.2. $S_D(k) = \sum_{s=1}^{D} M_{D,s}(k) = (2k+1)^D - (2k-1)^D$.

Justification. The classes in G.1 partition the shell. Equivalently, subtract the inner cube from the outer cube. The source assigns the origin its separate cardinality 1.

Source: III-A; (3); SRC-002.

G.3. The outward average is $u_k = (2k+1)^{D-1}/S_D(k)$.

Justification. Each point in class s has s outward directions among $2D$. Apply the following source-supplied binomial calculation to the average.

Source: III-C; (6)–(8); SRC-003.

G.3.1. $\sum_{s=1}^{D} s\binom{D}{s} 2^s a^{D-s} = 2D(a+2)^{D-1}$.

Justification. Differentiate the binomial expansion of $(a+2)^D$, multiply the derivative by a, and subtract it from $D(a+2)^D$. This is a polynomial identity, including $D = 1$.

Source: III-C; (7); SRC-004.

G.3.2. With $a = 2k - 1$, division by $2DS_D(k)$ gives the claim G.3.

Justification. Substitute G.3.1 into the average in (6); $D \geq 1$ and $S_D(k) > 0$ justify division.

Source: III-C; (8); SRC-005.

G.4. The inward average is $v_k = (2k-1)^{D-1}/S_D(k)$.

Justification. Only $s = 1$ permits a move to shell $k - 1$. Such a point has one inward direction. Use $M_{D,1}(k) = 2D(2k-1)^{D-1}$ and divide by $2DS_D(k)$.

Source: III-C; (9); SRC-006.

G.5. The quotient probabilities are pu_k, v_k, and $r_k = 1 - pu_k - v_k$.

Justification. Outward proposals are accepted with probability p; inward proposals are accepted with probability one. All remaining attempted moves leave the shell unchanged. These probabilities are an average over a finite shell, so the residual is nonnegative.

Source: III-C; after (9); SRC-007.

G.6. Q.E.D. for the stated shell cardinalities and proposal averages.

Justification. By G.2–G.5. These are averaged coefficients, not an assertion of Markov lumpability.

Source: III-C; end of calculation; SRC-008.

4.2 Theorem 1: exact quotient escape time

ASSUME / PROVE. Fix integers $D, N \geq 1$, $0 < p \leq 1$, and $\alpha = 1/p$. Let T_k be the quotient-chain mean hitting time of N, and $Q_D(N) = T_0$. Then

$$Q_D(N) = \sum_{\ell=0}^{N-1} \frac{\alpha^{\ell+1} + \sum_{m=1}^{\ell} S_D(m)\alpha^{\ell-m+1}}{(2\ell+1)^{D-1}}.$$

Source: IV-A–B; Eqs. (10)–(17). **Forward:** PASS WITH MINOR ISSUES. **Reverse:** FOLLOWS.

T1.1. Define $b_k = T_k - T_{k+1}$ for $0 \leq k < N$. Then $b_0 = 1/p$ and $pu_k b_k = v_k b_{k-1} + 1$ for $1 \leq k < N$, with $T_N = 0$.

Justification. Use the source first-step equations $T_k = 1 + pu_k T_{k+1} + v_k T_{k-1} + r_k T_k$ and $T_0 = 1 + pT_1 + (1-p)T_0$, and rearrange. Finite-mean absorption is an implicit finite-chain prerequisite; see the audit note A-01.

Source: IV-A; (10)–(12); SRC-009.

T1.2. For $R_k = v_k/(pu_k) = \alpha((2k-1)/(2k+1))^{D-1}$,

$$b_k = \left(\prod_{j=1}^{k} R_j\right) b_0 + \sum_{m=1}^{k} \frac{1}{pu_m} \prod_{j=m+1}^{k} R_j.$$

Justification. The source unrolls the recurrence for $k = 1, 2, 3$ and invokes induction for the displayed formula. The induction is on the transient index k; empty products are one and empty sums zero.

Source: IV-A; (13)–(14); SRC-010.

T1.3. For $0 \leq m < k$, $\prod_{j=m+1}^{k} R_j = \alpha^{k-m}((2m+1)/(2k+1))^{D-1}$.

Justification. The factors $(2j-1)/(2j+1)$ cancel successively. The source states the formula for $k \geq m+1$; its use at $k = m$ in the sum is the empty-product convention, noted in A-01.

Source: IV-B; (15); SRC-011.

T1.4. The initial contribution is $\alpha^{k+1}/(2k+1)^{D-1}$; the contribution indexed by $m \geq 1$ is $S_D(m)\alpha^{k-m+1}/(2k+1)^{D-1}$.

Justification. Insert $b_0 = 1/p$ and $1/(pu_m) = S_D(m)/(p(2m+1)^{D-1})$ into T1.2. Use T1.3 to cancel the positive factors.

Source: IV-B; terms preceding (16); SRC-012.

T1.5. $b_k = (2k+1)^{-(D-1)}[\alpha^{k+1} + \sum_{m=1}^{k} S_D(m)\alpha^{k-m+1}]$.

Justification. Add the two contributions in T1.4. This also covers $k = 0$.

Source: IV-B; (16); SRC-013.

T1.6. Q.E.D.: $Q_D(N) = \sum_{k=0}^{N-1} b_k$ has the asserted formula.

Justification. Telescope $T_k - T_{k+1}$ and use $T_N = 0$. For $N = 1$ the sum is $b_0 = 1/p$; no interior recurrence is needed.

Source: IV-B; (17); SRC-014.

4.3 Special cases and Remark 2

ASSUME / PROVE. The quotient formula specializes to $S_1(m) = 2$, $S_2(m) = 8m$, $S_3(m) = 24m^2+2$ for $m \geq 1$. For $p = 1$, $Q_D(N) = N^2$.

Source: IV-C; Eqs. (18)–(20), Remark 2. **Forward:** PASS. **Reverse:** FOLLOWS.

S.1. The three special-case formulas are obtained by replacing $S_D(m)$ by 2, $8m$, or $24m^2+2$ and the denominator by 1, $2\ell+1$, or $(2\ell+1)^2$.

Justification. Expand the powers in G and substitute into Theorem 1. This is an algebraic specialization, not an independent numerical validation.

Source: IV-C; (18)–(20); SRC-015.

S.2. If $p = 1$, then $1 + \sum_{m=1}^{\ell} S_D(m) = (2\ell+1)^D$ and $b_\ell = 2\ell+1$.

Justification. The successive odd powers cancel in the shell sum. Substitute $\alpha = 1$ into T1.

Source: IV-C; Remark 2; SRC-016.

S.3. Q.E.D.: $Q_D(N) = \sum_{\ell=0}^{N-1}(2\ell+1) = N^2$.

Justification. Sum the first N odd integers. Here $T_0^{(D)}$ is the quotient mean defined in Section IV; the identity does not assert dimension-independent true-chain escape times.

Source: IV-C; Remark 2; SRC-017.

4.4 Proposition 1: reversible shell quotient

ASSUME / PROVE. On $B_N = \{x : \|x\|_\infty \leq N\}$, reject proposals leaving the cube. The resulting un-killed chain is reversible with measure $\mu(x) = p^{\|x\|_\infty}$. Shorting shells while retaining shell masses gives transitions pu_k, v_k for $1 \leq k < N$ and outward probability p at 0. Stop either chain only upon hitting A_N.

Source: V-A; Eqs. (21)–(22). **Forward:** PASS. **Reverse:** FOLLOWS.

P1.1. $\mu(x)P(x,y) = \mu(y)P(y,x)$ for all pairs.

Justification. If $V(y) = V(x)+1$, both sides equal $p^{V(y)}/(2D)$. For equal-energy neighbors the proposal probabilities and weights agree. The reverse case follows by symmetry; nonedges have zero off-diagonal transition probabilities and self-loops balance themselves.

Source: V-A; detailed balance; SRC-018.

P1.2. The mass of A_k is $S_D(k)p^k$, and the cut conductance is $C_k = p^{k+1}(2k+1)^{D-1}$.

Justification. All vertices on A_k have weight p^k. There are $2D(2k+1)^{D-1}$ oriented outward edges, each with conductance $p^{k+1}/(2D)$. The origin is handled in P1.4.

Source: V-A; shell mass and edge count; SRC-019.

P1.3. For $1 \le k < N$, $\widetilde{P}(k,k+1) = C_k/(S_D(k)p^k) = pu_k$ and $\widetilde{P}(k,k-1) = C_{k-1}/(S_D(k)p^k) = v_k$.

Justification. Divide the retained intershell conductance by the retained shell mass, using $p > 0$ and $S_D(k) > 0$.

Source: V-A; after (22); SRC-020.

P1.4. $C_0 = p$ and $\widetilde{P}(0,1) = p$; internal moves become a self-loop.

Justification. The $2D$ edges from the origin have conductance $p/(2D)$ and the origin has mass one. Retaining all shell mass retains rejected proposals and internal transitions in the diagonal.

Source: V-A; final proof paragraph; SRC-021.

P1.5. Q.E.D. for reversibility and the stopped quotient construction.

Justification. Use P1.1–P1.4. Reflection outside B_N changes no transition before reaching its boundary. Absorption itself is not claimed reversible.

Source: V-A; proposition statement and close; SRC-022.

4.5 Theorem 2: quotient asymptotic order

> **ASSUME / PROVE.** For fixed integer $D \ge 1$ and fixed $0 < p < 1$, with $\alpha = 1/p > 1$, $Q_D(N) = \Theta(\alpha^N/N^{D-1})$ as $N \to \infty$.

Source: V-B; Eqs. (23)–(24). **Forward:** PASS WITH MINOR ISSUES. **Reverse:** FOLLOWS.

T2.1. Define $F_\ell = 1 + \sum_{m=1}^{\ell} S_D(m)\alpha^{-m}$. Then $b_\ell = \alpha^{\ell+1}F_\ell/(2\ell+1)^{D-1}$.

Justification. Factor $\alpha^{\ell+1}$ from the exact expression.

Source: V-B; (24); SRC-023.

T2.2. $1 \le F_\ell \uparrow F_\infty < \infty$, and $b_\ell \asymp \alpha^\ell/\ell^{D-1}$ as $\ell \to \infty$.

Justification. The shell polynomial is of order m^{D-1}, and its product with α^{-m} is summable because $\alpha > 1$. All summands are nonnegative. The comparison concerns large positive ℓ, not $\ell = 0$.

Source: V-B; convergence paragraph; SRC-024.

T2.3. $b_{\ell-1}/b_\ell = \alpha^{-1}((2\ell+1)/(2\ell-1))^{D-1}F_{\ell-1}/F_\ell \to \alpha^{-1} < 1$.

Justification. Use the factorization in T2.1, the positive finite limit from T2.2, and elementary limits.

Source: V-B; ratio display; SRC-025.

T2.4. There are $q \in (0,1)$ and L with $b_{\ell-1} \le qb_\ell$ for $\ell \ge L$, and $b_{N-1} \le Q_D(N) \le C'b_{N-1}$ for all sufficiently large N.

Justification. The ratio limit supplies q, L. The source bounds the tail by $b_{N-1}/(1-q)$ and absorbs the fixed prefix $\sum_{\ell=0}^{L-1} b_\ell$ into $C'b_{N-1}$. Eventual growth and the qualification "sufficiently large are implicit in that last absorption; see A-02.

Source: V-B; geometric domination display; SRC-026.

T2.5. Q.E.D.: $Q_D(N) = \Theta(\alpha^N / N^{D-1})$.

Justification. $b_{N-1} = \alpha^N F_{N-1}/(2N-1)^{D-1}$ has this order by T2.2. Combine with T2.4. Constants may depend on fixed D, p.

Source: V-B; final sentence; SRC-027.

4.6 Remark 3: sharper leading constant

ASSUME / PROVE. Under the assumptions of Theorem 2, the paper claims $Q_D(N) \sim \frac{\alpha}{\alpha-1} F_\infty \frac{\alpha^N}{(2N)^{D-1}}$.

Source: V-B; Eq. (25). **Forward:** PASS. **Reverse:** FOLLOWS.

R3.1. Put $r = \alpha^{-1} \in (0,1)$. For each fixed integer $j \geq 0$, $b_{N-1-j}/b_{N-1} \to r^j$.

Justification. For $N > j$, express the ratio as the product of j consecutive backward ratios. Each tends to r by Theorem 2. The empty product at $j = 0$ is one.

Source: V-B; Remark 3; eq:sharp-asymptotic and added limit argument; SRC-028.

R3.2. Choose $r < q < 1$ and L so that $b_{\ell-1} \leq q b_\ell$ for all $\ell \geq L$. For $N - 1 - j \geq L$, $b_{N-1-j}/b_{N-1} \leq q^j$.

Justification. The backward ratio limit gives this threshold. Repeatedly apply the inequality through the finite product of ratios. All b_ℓ are positive.

Source: V-B; Remark 3; eq:sharp-asymptotic and added limit argument; SRC-029.

R3.3. $Q_D(N)/b_{N-1} \to \sum_{j=0}^{\infty} r^j = \alpha/(\alpha-1)$.

Justification. Read the sum from $N-1$ backward. Set terms outside $0 \leq j \leq N-1-L$ to zero. For each fixed j, they eventually have the limit in R3.1; R3.2 bounds them by the summable sequence q^j. Dominated convergence for a series gives the tail limit. The fixed prefix $\sum_{\ell<L} b_\ell/b_{N-1}$ vanishes because $b_{N-1} \to \infty$, as supplied by the factorization and $\alpha > 1$.

Source: V-B; Remark 3; eq:sharp-asymptotic and added limit argument; SRC-030.

R3.4. $Q_D(N) \sim \frac{\alpha}{\alpha-1} F_\infty \frac{\alpha^N}{(2N)^{D-1}}$.

Justification. Use $b_{N-1} = \alpha^N F_{N-1}/(2N-1)^{D-1}$, $F_{N-1} \to F_\infty > 0$, and $(2N-1)/(2N) \to 1$, with fixed D. Combine with R3.3.

Source: V-B; Remark 3; eq:sharp-asymptotic and added limit argument; SRC-031.

R3.5. Q.E.D. for Remark 3.

Justification. R3.4 is the claimed sharper equivalent. The former GAP-001 is closed by the explicit added source argument.

Source: V-B; Remark 3; eq:sharp-asymptotic and added limit argument; SRC-032.

4.7 Lemma 1: hitting time via capacity

ASSUME / PROVE. Fix $D \geq 1$, $0 < p < 1$, a nonempty finite $A \subset \mathbb{Z}^D$, and $a \notin A$. For the un-killed chain, put $h(x) = \Pr_x(\tau_a < \tau_A)$ and $M = \sum_x \mu(x)$. The claims are

$$\mathbb{E}_a \tau_A = \frac{\sum_x \mu(x) h(x)}{\mathrm{Cap}(a, A)}, \qquad \mu(a) R_{\mathrm{eff}}(a, A) \leq \mathbb{E}_a \tau_A \leq M R_{\mathrm{eff}}(a, A).$$

Source: V-C; Eqs. (26)–(27). **Forward:** INCOMPLETE. **Reverse:** INDETERMINATE FROM THE PROVIDED MATERIAL.

L1.1. $M = 1+\sum_{k\geq 1} S_D(k)p^k < \infty$; the irreducible chain has stationary probability μ/M and is positive recurrent. The source concludes that the fixed-state hitting and return times used below have finite means.

Justification. **EXT-001; EXTERNAL UNAVAILABLE**. The mass series converges by polynomial–geometric summability. The invoked countable-chain recurrence and hitting-time background is retained as EXT-001; the exact external theorem statement is not supplied.

Source: V-C; proof opening; SRC-033.

L1.O1. OBLIGATION: obtain the exact countable-state recurrence/hitting-time result invoked by L1.1 and verify it for this irreducible chain with invariant probability μ/M.

Justification. **EXT-001; EXTERNAL UNAVAILABLE**. External text was not supplied or reverified in this update. This row records unavailable support, not an assertion that the invoked result is false.

Source: V-C; proof opening; SRC-034.

L1.2. For $h(x) = \Pr_x(\tau_a < \tau_A)$, $h(a) = 1$, $h|_A = 0$, and $Ph = h$ off $\{a\} \cup A$. For bounded functions,

$$\mathcal{E}(f, g) = \sum_x \mu(x)g(x)(f(x) - Pf(x)).$$

Justification. Condition on the first step for harmonicity. Expand the symmetric mixed energy and use detailed balance. Boundedness and $\sum_{x,y} \mu(x)P(x, y) = M < \infty$ justify absolute convergence. Self-loops give zero energy.

Source: V-C; harmonicity and mixed energy identity; SRC-035.

L1.3. With $q = \Pr_a(\tau_A < \tau_a^+)$, $\mathcal{E}(h, h) = \mu(a)(1 - Ph(a)) = \mu(a)q$.

Justification. In L1.2, only the contribution at a survives. The first-step return decomposition gives $1 - Ph(a) = q$. Recurrence excludes avoiding both boundary sets forever. Rejected proposals count as returns at time one.

Source: V-C; capacity computation; SRC-036.

L1.4. h minimizes energy over $f(a) = 1$, $f|_A = 0$.

Justification. The source first clips f into $[0, 1]$, which preserves its boundary values and cannot increase energy. Write the clipped $f = h + g$. Harmonicity and $g|_{\{a\}\cup A} = 0$ give $\mathcal{E}(h, g) = 0$. Hence $\mathcal{E}(f, f) = \mathcal{E}(h, h) + \mathcal{E}(g, g) \geq \mathcal{E}(h, h)$. All functions in this expansion are bounded.

Source: V-C; energy minimization; SRC-037.

L1.5. $\mathrm{Cap}(a, A) = \mu(a)q$.

Justification. Use the attained variational minimum in L1.4 and its value in L1.3. A simple finite path from a into A, without a prior return to a, has positive probability, so $q > 0$. The positivity check is implicit in irreducibility.

Source: V-C; end of capacity argument; SRC-038.

L1.6. For $G_A(u, x) = \mathbb{E}_u \sum_{t<\tau_A} \mathbf{1}_{\{X_t=x\}}$,

$$\mathbb{E}_a\tau_A = \sum_x G_A(a, x), \qquad \mu(a)G_A(a, x) = \mu(x)G_A(x, a).$$

Justification. Sum the nonnegative occupation counts. Apply detailed balance to each finite path avoiding A, reverse the path, and sum over lengths. This is reversibility of the killed Green function, not of an absorbing transition kernel with positive target weights.

Source: V-C; occupation sum and killed reversibility; SRC-039.

L1.7. $G_A(x, a) = h(x)G_A(a, a)$ and $G_A(a, a) = 1/q = \mu(a)/\mathrm{Cap}(a, A)$.

Justification. Use the strong Markov property at the first hit of a before A. Starting at a, each visit has probability q of escape before the next return; the number of visits, including the initial one, is geometric. This argument is explicit in the latest source.

Source: V-C; strong Markov property and geometric visits; SRC-040.

L1.8. $\mathbb{E}_a \tau_A = \big(\sum_x \mu(x) h(x)\big) / \operatorname{Cap}(a, A)$.

Justification. Substitute the two Green identities from L1.6–L1.7 into the occupation sum and cancel the positive $\mu(a)$.

Source: V-C; Eq. (26); SRC-041.

L1.9. $R_{\text{eff}}(a, A) = 1/\operatorname{Cap}(a, A)$, so $\mu(a) R_{\text{eff}} \le \mathbb{E}_a \tau_A \le M R_{\text{eff}}$.

Justification. **EXT-002; EXTERNAL UNAVAILABLE**. The source invokes the network principle with its variational capacity convention. Its exact infinite-network, finite-target statement remains EXT-002. Bound the numerator in L1.8 using $h(a) = 1$ and $0 \le h \le 1$.

Source: V-C; Eqs. (26)–(27); SRC-042.

L1.O2. OBLIGATION: obtain the finite-target, countably infinite network capacity/resistance duality and unit-flow inequality in the source convention, and check their hypotheses.

Justification. **EXT-002; EXTERNAL UNAVAILABLE**. External text was not supplied or reverified in this update. This row records unavailable support, not an assertion that the invoked result is false.

Source: V-C; capacity convention; V-E; Thomson application; SRC-043.

L1.10. Q.E.D. for Lemma 1, conditional on EXT-001 and EXT-002.

Justification. L1.8 proves the identity; L1.9 gives both inequalities.

Source: V-C; conclusion; SRC-044.

4.8 Theorem 3: true-chain lower bound

ASSUME / PROVE. For fixed integer $D \ge 1$, $0 < p < 1$, and every integer $N \ge 1$,

$$H_D(N) \ge \frac{\alpha^N}{(2N-1)^{D-1}} \ge 2^{-(D-1)} \frac{\alpha^N}{N^{D-1}}.$$

Source: V-D; Eq. (28). **Forward:** INCOMPLETE. **Reverse:** INDETERMINATE FROM THE PROVIDED MATERIAL.

T3.1. The exterior of B_N does not change the resistance from 0 to A_N.

Justification. The shell separates 0 from the exterior; an equilibrium potential is zero outside. Thus shorting may be performed on the finite cube. This is the source restriction argument.

Source: V-D; separation argument; SRC-045.

T3.2. $R_{\text{eff}}(0, A_N) \ge R_{\text{eff}}^{\text{coll}}(0, N)$.

Justification. Short all vertices of each shell. Rayleigh's shorting principle applies to this finite network with positive conductances; the source cites Doyle–Snell and Levin–Peres.

Source: V-D; Rayleigh monotonicity; SRC-046.

T3.3. $R_{\text{eff}}^{\text{coll}}(0, N) = \sum_{k=0}^{N-1} \alpha^{k+1}/(2k+1)^{D-1}$, of order α^N / N^{D-1}.

Justification. The intershell conductances from P1 are in series. The ratio/geometric-tail calculation is the one used in Theorem 2 with the factor F_ℓ absent. The next source step supplies an alternative lower estimate requiring no asymptotic summation.

Source: V-D; resistance sum; SRC-047.

T3.4. $H_D(N) \geq R_{\text{eff}}(0, A_N) \geq \alpha^N/(2N-1)^{D-1}$.

Justification. Apply the lower inequality of Lemma 1 with $a = 0$, where $\mu(0) = 1$. In the positive series of T3.3 retain only $k = N - 1$.

Source: V-D; last-term refinement; SRC-048.

T3.5. Q.E.D. with $c_{D,p} = 2^{-(D-1)}$.

Justification. Since $2N - 1 \leq 2N$ and $D - 1 \geq 0$, T3.4 implies the asserted inequality for every $N \geq 1$. Its dependence on the unresolved background of L1 remains explicit.

Source: V-D; last display; SRC-049.

4.9 Lemma 2: transport on nested cubes

> **ASSUME / PROVE.** For integers $n \geq 1$, $k \geq 0$, let $Q_k = \{-k, \ldots, k\}^n$, ρ_k be its one-dimensional uniform law extended by zero, and $u_k = \rho_k^{\otimes n}$. There is a nearest-neighbor flow ψ_k on Q_{k+1} such that $\operatorname{div} \psi_k = u_k - u_{k+1}$ and $\sum_e \psi_k(e)^2 \leq n/(k+1)^n$, counting each unoriented edge once.

Source: V-E; Eq. (29). **Forward:** PASS. **Reverse:** FOLLOWS.

L2.1. For $n = 1$, orient edges $j \to j + 1$ and set $J_{j+1/2} = \sum_{t=-k-1}^{j} (\rho_k(t) - \rho_{k+1}(t))$ for $-k - 1 \leq j \leq k$, with flux zero outside.

Justification. This is the source definition. It specifies signed flow on every oriented edge, with the reverse orientation its negative.

Source: V-E; one-dimensional construction; SRC-050.

L2.2. $J_{j+1/2} - J_{j-1/2} = \rho_k(j) - \rho_{k+1}(j)$, including the endpoints, and

$$J_{j+1/2} = -\frac{1}{2k+3} + \frac{2(j+k+1)}{(2k+1)(2k+3)}.$$

Justification. Telescope the cumulative sums for divergence. For the displayed flux, count the interior terms in the sum of L2.1. The endpoint convention uses that the total difference of the two probability masses is zero.

Source: V-E; divergence and explicit evaluation; SRC-051.

L2.3. $|J_{j+1/2}| \leq 1/(2k+3)$ and $\sum_{j=-k-1}^{k} J_{j+1/2}^2 \leq (2k+2)/(2k+3)^2 \leq 1/(k+1)$.

Justification. The explicit affine formula in L2.2 ranges between the two endpoint values $-1/(2k+3)$ and $1/(2k+3)$. There are $2k + 2$ edges. All denominators are positive, including at $k = 0$.

Source: V-E; one-dimensional energy bound; SRC-052.

L2.4. Define $\nu^{(r)} = \rho_{k+1}^{\otimes r} \otimes \rho_k^{\otimes(n-r)}$, $0 \leq r \leq n$. For stage $r < n$, the flow in coordinate $r + 1$ is $w_r(z) J_{j+1/2}$, where $w_r(z) = \prod_{i=1}^{r} \rho_{k+1}(z_i) \prod_{i=r+2}^{n} \rho_k(z_i)$.

Justification. The source changes one coordinate at a time. Other directions have zero stage flow. Empty products equal one.

Source: V-E; product-stage construction; SRC-053.

L2.5. $\operatorname{div} \psi^{(r)} = \nu^{(r)} - \nu^{(r+1)}$.

Justification. On a fixed fiber, multiply the divergence identity L2.2 by the constant weight $w_r(z)$. The product laws differ only in the active coordinate.

Source: V-E; fiber divergence; SRC-054.

L2.6. $\sum_z w_r(z)^2 = (2k+3)^{-r}(2k+1)^{-(n-r-1)} \le (k+1)^{-(n-1)}$.

Justification. Factor the finite sum of products. A uniform law on m points has sum of squares $1/m$. There are $n-1$ fixed coordinates; $2k+1, 2k+3 \ge k+1$.

Source: V-E; squared fiber masses; SRC-055.

L2.7. $\sum_e \psi^{(r)}(e)^2 = (\sum_z w_r(z)^2)(\sum_j J_{j+1/2}^2) \le (k+1)^{-n}$.

Justification. The fiber index and edge index separate, so use L2.3 and L2.6.

Source: V-E; stage energy; SRC-056.

L2.8. For $\psi_k = \sum_{r=0}^{n-1} \psi^{(r)}$, the divergence is $u_k - u_{k+1}$ and the energy is at most $n/(k+1)^n$.

Justification. The divergences in L2.5 telescope between $\nu^{(0)} = u_k$ and $\nu^{(n)} = u_{k+1}$. Distinct stages have distinct coordinate directions and therefore disjoint unoriented edge sets. Their squared energies add without cross terms.

Source: V-E; summing stage flows; SRC-057.

L2.9. Q.E.D. with $C_n = n$, including $k = 0$.

Justification. Use the explicitly constructed flow L2.8; no limiting argument or external network theorem is required.

Source: V-E; end of transport proof; SRC-058.

4.10 Proposition 2: resistance upper bound

> **ASSUME / PROVE.** For fixed $D \ge 1$, $0 < p < 1$, let $F_N^\circ = \{(N, y) : |y_i| \le N-1\}$. There is $C'_{D,p} < \infty$, independent of integer $N \ge 1$, with $R_{\text{eff}}(0, F_N^\circ) \le C'_{D,p}\alpha^N/N^{D-1}$.

Source: V-E; Eq. (30). **Forward:** INCOMPLETE. **Reverse:** INDETERMINATE FROM THE PROVIDED MATERIAL.

P2.1. CASE $D = 1$: unit current on the positive path has energy $2\sum_{k=0}^{N-1} \alpha^{k+1} \le 2\alpha^{N+1}/(\alpha-1)$.

Justification. Each path edge has conductance $p^{k+1}/2$. Thomson's inequality gives the resistance upper bound. The factor two from the revised source is retained. The infinite-network application is covered by EXT-002.

Source: V-E; D=1 correction; SRC-059.

P2.2. CASE $D \ge 2$: put $n = D-1$; use layers $L_k = \{(k, y) : y \in Q_k\}$, $0 \le k < N$, and target $L_N^\circ = \{(N, y) : y \in Q_{N-1}\}$.

Justification. The source builds the following unit flow. At each departure layer L_k the outgoing distribution is uniform on Q_k.

Source: V-E; D>=2 construction; SRC-060.

> **P2.2.1.** Each of the $(2k+1)^n$ vertical edges carries $(2k+1)^{-n}$, with energy $E_k^{\text{vert}} = 2D\alpha^{k+1}/(2k+1)^n \le 2D\alpha^{k+1}/(k+1)^n$.
>
> *Justification.* For such edges the energy increases from k to $k+1$, so the conductance is $p^{k+1}/(2D)$. Multiply the squared current divided by conductance by the number of edges.
>
> Source: V-E; forward step; SRC-061.

> **P2.2.2.** For $0 \le k \le N-2$, redistribute u_k to u_{k+1} within L_{k+1} at energy $E_{k+1}^{\text{hor}} \le 2Dn\alpha^{k+1}/(k+1)^n$.
>
> *Justification.* Apply Lemma 2 and multiply its unit-conductance cost by $2D\alpha^{k+1}$. All vertices of this layer have energy $k+1$, so each lateral conductance is $p^{k+1}/(2D)$.
>
> Source: V-E; lateral redistribution; SRC-062.

P2.2.3. At intermediate L_j, divergence is $-u_{j-1} + u_j + (u_{j-1} - u_j) = 0$; source divergence is 1 and total target divergence is -1.

Justification. Incoming vertical current contributes the negative term; outgoing vertical current the positive term; Lemma 2 gives lateral divergence. Extend u_{j-1} by zero outside Q_{j-1}. At the last step there is no redistribution. At $N = 1$ there is only one vertical edge.

Source: V-E; Conservation of current; SRC-063.

P2.2.4. After extension by zero, the flow is feasible on $\mathbb{Z}^D$ and

$$R_{\text{eff}}(0, F_N^\circ) \le \sum_{k=0}^{N-1} E_k^{\text{vert}} + \sum_{k=1}^{N-1} E_k^{\text{hor}} \le C_{D,p} \sum_{k=0}^{N-1} \frac{\alpha^{k+1}}{(k+1)^n}.$$

Justification. Vertical edges and lateral edges in distinct layers are disjoint. Add their energies and apply Thomson's principle, subject to EXT-002.

Source: V-E; energy sum and Thomson; SRC-064.

P2.2.5. For $a_j = \alpha^j / j^n$, choose $q \in (\alpha^{-1}, 1)$ and J with $a_{j-1} \le q a_j$ for $j \ge J + 1$. Then $\sum_{j=1}^N a_j \le K_{n,p} a_N$ for all $N \ge 1$.

Justification. The ratio tends to α^{-1}. For $N \ge J$, bound the tail by $a_N/(1-q)$ and absorb the finite prefix using $a_N \ge a_J > 0$. Enlarge the constant for the finitely many $1 \le N < J$. This entire argument is present in the revised source.

Source: V-E; ratio argument; SRC-065.

P2.2.6. Q.E.D. for the $D \ge 2$ case, conditional on EXT-002.

Justification. Combine P2.2.4 with P2.2.5 and $n = D - 1$. Export the completed case as P2.2.

Source: V-E; end of D>=2 case; SRC-066.

P2.3. Q.E.D. for Proposition 2.

Justification. The cases $D = 1$ and $D \ge 2$ exhaust integer $D \ge 1$. Use the completed cases P2.1 and P2.2. The constant depends on D, p, not N.

Source: V-E; end of proposition proof; SRC-067.

4.11 Theorem 4: true-chain asymptotic scaling

ASSUME / PROVE. For fixed integer $D \ge 1$, fixed $0 < p < 1$, and the true Metropolis chain started at 0, $H_D(N) = \Theta(\alpha^N / N^{D-1})$ as $N \to \infty$.

Source: V-E; Eq. (31). **Forward:** INCOMPLETE. **Reverse:** INDETERMINATE FROM THE PROVIDED MATERIAL.

T4.1. $H_D(N) \ge 2^{-(D-1)} \alpha^N / N^{D-1}$.

Justification. Invoke Theorem 3.

Source: Theorem 4 proof; opening; SRC-068.

T4.2. $H_D(N) \le \mathbb{E}_0 \tau_{F_N^\circ} \le M R_{\text{eff}}(0, F_N^\circ) \le M C'_{D,p} \alpha^N / N^{D-1}$.

Justification. $F_N^\circ \subset A_N$ gives the pathwise comparison of first hitting times. Apply Lemma 1 to the nonempty finite face, then Proposition 2.

Source: Theorem 4 proof; upper display; SRC-069.

T4.3. Q.E.D. for the claimed Θ law, conditional on the recorded background obligations.

Justification. The finite $M = M_{D,p}$ and the constants in T4.1–T4.2 depend only on fixed D, p. The positive upper and lower constants establish the claimed order.

Source: Theorem 4 proof; final sentence; SRC-070.

4.12 Ancillary symmetry and numerical-method identities

> **ASSUME / PROVE.** The shell quotient is generally not the exact shell dynamics, whereas grouping points by coordinate signs and order preserves the true dynamics. The active Section VI uses sparse escape-probability and excursion-time systems. The comparison to the earlier one-dimensional paper is separate external support.

Source: II; III-C; IV-C Remark 1; VI-A. **Forward:** INCOMPLETE. **Reverse:** INDETERMINATE FROM THE PROVIDED MATERIAL.

N.1. For $D \geq 2$, $1 \leq k < N$, $(k, 0, \ldots, 0)$ and $(k, k, 0, \ldots, 0)$ have outward proposal probabilities $1/(2D)$ and $1/D$.

Justification. Count the coordinates on the shell boundary. The accepted outward probabilities are $p/(2D)$ and p/D. Thus shell membership alone does not specify the transition probabilities; the shell quotient is a separate chain.

Source: III-C; shell counterexample; SRC-071.

N.2. Grouping lattice points that differ only in coordinate signs or coordinate order preserves the transition probabilities between groups and the escape-time calculation.

Justification. **EXT-003; EXTERNAL UNAVAILABLE**. The source invokes the Kemeny–Snell aggregation criterion and the kernel symmetry. EXT-003 records the exact external-text check. No group-theory notation is required.

Source: II; aggregation paragraph; VI-A; SRC-072.

N.O3. OBLIGATION: obtain Kemeny–Snell Sec. 6.3, Thm. 6.3.2 and verify its aggregation criterion for the stated symmetry groups.

Justification. **EXT-003; EXTERNAL UNAVAILABLE**. External text was not supplied or reverified in this update. This row records unavailable support, not an assertion that the invoked result is false.

Source: II; aggregation paragraph; SRC-073.

N.3. The true mean is computed using sparse systems for escape probabilities and excursion times on these symmetry-equivalent groups.

Justification. This is the active numerical-method description. The earlier explicit regenerative formula is now commented out in the source; it is not treated as a current displayed claim. Numerical accuracy and plotted speedups are outside this proof audit.

Source: VI-A; Numerical Method; SRC-074.

N.4. The $D = 1$ quotient formula matches the predecessor after the parameter identification $p = 2p_{\text{prior}}$.

Justification. **EXT-004; EXTERNAL UNAVAILABLE**. The source cites its predecessor. The predecessor's exact parameter convention and formula were not included in this audit (EXT-004). This comparison is not used to establish Theorems 1–4.

Source: IV-C; Remark 1; SRC-075.

N.O4. OBLIGATION: obtain the predecessor formula and parameter convention, and check $p = 2p_{\text{prior}}$.

Justification. **EXT-004; EXTERNAL UNAVAILABLE**. External text was not supplied or reverified in this update. This row records unavailable support, not an assertion that the invoked result is false.

Source: IV-C; Remark 1; SRC-076.

5 Source-to-step mapping ledger

Each contract maps to the source unit identified in the inventory. The rows below map every rendered step. Source IDs index the precise paraphrases printed at those steps; they are provenance, not extra premises. A repeated source paragraph may produce several normalized steps. Obligation rows preserve missing support and do not count as supplied proof evidence.

Step	Source ID	Mapping	Support	Issue	Notes
G contract	SRC-G-C	NORMALIZED	EXPLICIT	–	Contract
G.1	SRC-001	NORMALIZED	EXPLICIT	–	Same source location as step
G.2	SRC-002	NORMALIZED	EXPLICIT	–	Same source location as step
G.3	SRC-003	NORMALIZED	EXPLICIT	–	Same source location as step
G.3.1	SRC-004	NORMALIZED	EXPLICIT	–	Same source location as step
G.3.2	SRC-005	NORMALIZED	EXPLICIT	–	Same source location as step
G.4	SRC-006	NORMALIZED	EXPLICIT	–	Same source location as step
G.5	SRC-007	NORMALIZED	EXPLICIT	–	Same source location as step
G.6	SRC-008	STRUCTURAL	IMPLICIT	–	Same source location as step
T1 contract	SRC-T1-C	NORMALIZED	EXPLICIT	–	Contract
T1.1	SRC-009	NORMALIZED	EXPLICIT	–	Same source location as step
T1.2	SRC-010	NORMALIZED	EXPLICIT	–	Same source location as step
T1.3	SRC-011	NORMALIZED	EXPLICIT	–	Same source location as step
T1.4	SRC-012	NORMALIZED	EXPLICIT	–	Same source location as step
T1.5	SRC-013	NORMALIZED	EXPLICIT	–	Same source location as step
T1.6	SRC-014	STRUCTURAL	IMPLICIT	–	Same source location as step
S contract	SRC-S-C	NORMALIZED	EXPLICIT	–	Contract
S.1	SRC-015	NORMALIZED	EXPLICIT	–	Same source location as step
S.2	SRC-016	NORMALIZED	EXPLICIT	–	Same source location as step
S.3	SRC-017	STRUCTURAL	IMPLICIT	–	Same source location as step
P1 contract	SRC-P1-C	NORMALIZED	EXPLICIT	–	Contract
P1.1	SRC-018	NORMALIZED	EXPLICIT	–	Same source location as step
P1.2	SRC-019	NORMALIZED	EXPLICIT	–	Same source location as step
P1.3	SRC-020	NORMALIZED	EXPLICIT	–	Same source location as step
P1.4	SRC-021	NORMALIZED	EXPLICIT	–	Same source location as step
P1.5	SRC-022	STRUCTURAL	IMPLICIT	–	Same source location as step
T2 contract	SRC-T2-C	NORMALIZED	EXPLICIT	–	Contract
T2.1	SRC-023	NORMALIZED	EXPLICIT	–	Same source location as step
T2.2	SRC-024	NORMALIZED	EXPLICIT	–	Same source location as step
T2.3	SRC-025	NORMALIZED	EXPLICIT	–	Same source location as step
T2.4	SRC-026	NORMALIZED	EXPLICIT	–	Same source location as step
T2.5	SRC-027	STRUCTURAL	IMPLICIT	–	Same source location as step
R3 contract	SRC-R3-C	NORMALIZED	EXPLICIT	–	Contract

Step	Source ID	Mapping	Support	Issue	Notes
R3.1	SRC-028	NORMALIZED	EXPLICIT	–	Same source location as step
R3.2	SRC-029	NORMALIZED	EXPLICIT	–	Same source location as step
R3.3	SRC-030	NORMALIZED	EXPLICIT	–	Same source location as step
R3.4	SRC-031	NORMALIZED	EXPLICIT	–	Same source location as step
R3.5	SRC-032	STRUCTURAL	IMPLICIT	–	Same source location as step
L1 contract	SRC-L1-C	NORMALIZED	EXPLICIT	–	Contract
L1.1	SRC-033	NORMALIZED	EXTERNAL UNAVAILABLE	EXT-001	Same source location as step
L1.O1	SRC-034	OBLIGATION	EXTERNAL UNAVAILABLE	EXT-001	Same source location as step
L1.2	SRC-035	NORMALIZED	EXPLICIT	–	Same source location as step
L1.3	SRC-036	NORMALIZED	EXPLICIT	–	Same source location as step
L1.4	SRC-037	NORMALIZED	EXPLICIT	–	Same source location as step
L1.5	SRC-038	NORMALIZED	EXPLICIT	–	Same source location as step
L1.6	SRC-039	NORMALIZED	EXPLICIT	–	Same source location as step
L1.7	SRC-040	NORMALIZED	EXPLICIT	–	Same source location as step
L1.8	SRC-041	NORMALIZED	EXPLICIT	–	Same source location as step
L1.9	SRC-042	NORMALIZED	EXTERNAL UNAVAILABLE	EXT-002	Same source location as step
L1.O2	SRC-043	OBLIGATION	EXTERNAL UNAVAILABLE	EXT-002	Same source location as step
L1.10	SRC-044	STRUCTURAL	IMPLICIT	–	Same source location as step
T3 contract	SRC-T3-C	NORMALIZED	EXPLICIT	–	Contract
T3.1	SRC-045	NORMALIZED	EXPLICIT	–	Same source location as step
T3.2	SRC-046	NORMALIZED	EXPLICIT	–	Same source location as step
T3.3	SRC-047	NORMALIZED	EXPLICIT	–	Same source location as step
T3.4	SRC-048	NORMALIZED	EXPLICIT	–	Same source location as step
T3.5	SRC-049	STRUCTURAL	IMPLICIT	–	Same source location as step
L2 contract	SRC-L2-C	NORMALIZED	EXPLICIT	–	Contract
L2.1	SRC-050	NORMALIZED	EXPLICIT	–	Same source location as step
L2.2	SRC-051	NORMALIZED	EXPLICIT	–	Same source location as step
L2.3	SRC-052	NORMALIZED	EXPLICIT	–	Same source location as step
L2.4	SRC-053	NORMALIZED	EXPLICIT	–	Same source location as step
L2.5	SRC-054	NORMALIZED	EXPLICIT	–	Same source location as step
L2.6	SRC-055	NORMALIZED	EXPLICIT	–	Same source location as step
L2.7	SRC-056	NORMALIZED	EXPLICIT	–	Same source location as step
L2.8	SRC-057	NORMALIZED	EXPLICIT	–	Same source location as step
L2.9	SRC-058	STRUCTURAL	IMPLICIT	–	Same source location as step
P2 contract	SRC-P2-C	NORMALIZED	EXPLICIT	–	Contract
P2.1	SRC-059	NORMALIZED	EXPLICIT	–	Same source location as step
P2.2	SRC-060	NORMALIZED	EXPLICIT	–	Same source location as step

Step	Source ID	Mapping	Support	Issue	Notes
P2.2.1	SRC-061	NORMALIZED	EXPLICIT	–	Same source location as step
P2.2.2	SRC-062	NORMALIZED	EXPLICIT	–	Same source location as step
P2.2.3	SRC-063	NORMALIZED	EXPLICIT	–	Same source location as step
P2.2.4	SRC-064	NORMALIZED	EXPLICIT	–	Same source location as step
P2.2.5	SRC-065	NORMALIZED	EXPLICIT	–	Same source location as step
P2.2.6	SRC-066	STRUCTURAL	IMPLICIT	–	Same source location as step
P2.3	SRC-067	STRUCTURAL	IMPLICIT	–	Same source location as step
T4 contract	SRC-T4-C	NORMALIZED	EXPLICIT	–	Contract
T4.1	SRC-068	NORMALIZED	EXPLICIT	–	Same source location as step
T4.2	SRC-069	NORMALIZED	EXPLICIT	–	Same source location as step
T4.3	SRC-070	STRUCTURAL	IMPLICIT	–	Same source location as step
N contract	SRC-N-C	NORMALIZED	EXPLICIT	–	Contract
N.1	SRC-071	NORMALIZED	EXPLICIT	–	Same source location as step
N.2	SRC-072	NORMALIZED	EXTERNAL UNAVAILABLE	EXT-003	Same source location as step
N.O3	SRC-073	OBLIGATION	EXTERNAL UNAVAILABLE	EXT-003	Same source location as step
N.3	SRC-074	NORMALIZED	EXPLICIT	–	Same source location as step
N.4	SRC-075	NORMALIZED	EXTERNAL UNAVAILABLE	EXT-004	Same source location as step
N.O4	SRC-076	OBLIGATION	EXTERNAL UNAVAILABLE	EXT-004	Same source location as step

6 Gap and external-support register

GAP-001 — CLOSED in this revision. Remark 3 now proves

$$Q_D(N)/b_{N-1} \longrightarrow \alpha/(\alpha-1).$$

Steps R3.1–R3.3 give pointwise limits and geometric domination of the backward terms, together with disappearance of the fixed prefix. R3.4 converts this limit to the stated sharper equivalent. This closure applies to the quotient refinement and does not change the dependency route for the main true-chain theorem.

EXT-001 — countable-state background. Obtain the exact invoked theorem statements and check that irreducibility plus the invariant probability μ/M supplies recurrence and the fixed-state finite hitting/return means used in L1.1. The source's polynomial–geometric mass estimate is checked. Unavailable support propagates through L1, T3 and T4. The explicit meta-obligation is L1.O1.

EXT-002 — finite-target infinite-network convention. Obtain and check the exact capacity/resistance duality and the unit-flow inequality for the countable network with the variational capacity defined above. Match finite targets, boundary behavior, normalization and self-loops. L1.O2 records this obligation; it affects L1.9, the network application in P2, and T3/T4. The finite-support flow and its energy are locally checked independently of this external boundary.

EXT-003 — aggregation criterion. Obtain Kemeny–Snell Sec. 6.3, Thm. 6.3.2 and verify its exact applicability to the grouping by coordinate signs/order. N.O3 records this obligation; it affects N.2–N.3, not the main analytical theorem.

EXT-004 — predecessor comparison. Obtain the cited predecessor's exact formula and proposal/acceptance convention to verify $p = 2p_{\text{prior}}$. N.O4 records the obligation; only N.4 depends on it.

Editorial changes applied. Equation `eq:sharp-asymptotic` now has ordinary multiplication spacing and the

full displayed factor. The bias-ratio definition permits $\alpha = 1$, while its strict inequality is restricted to $p < 1$. The repository paragraph names the numerical notebook and Lamport-proof supplement; the authors will upload the latter. The revised acknowledgment names ChatGPT, Claude and Grok and retains author responsibility.

7 Frozen audit handoff

The contracts, rendering, source identity, step labels, support statuses and register above form the frozen handoff. The audits below evaluate that rendering with GAP-001 explicitly closed by new source text, while unavailable external support remains open. Mapping metadata is not proof evidence. Any future correction should change the source hash and identify which obligations it closes.

Part II

Forward audit

8 Verdict and theorem contracts

Overall verdict: INCOMPLETE under the frozen external-source boundary. The previous internal gap in Remark 3 is closed. The exact quotient formula, its asymptotic order, the sharper equivalent, and the transport construction close with at most minor exposition clarifications. The true-chain proof still invokes the recorded countable-state and network results whose exact external statements were not reverified here. The four theorem contracts are T1, T2, T3 and T4; limits keep D, p fixed.

9 Proof structure and findings

The rendering contains twelve units and two numbered levels within the constructive proofs. Source order is preserved within each mathematical argument. Lemma 1 now proves the capacity identity before using it in the Green-function hitting-time formula; the previous supplement's provisional order and descriptions have been updated.

RESOLVED — R3.1–R3.5. The new source gives the formerly missing leading-constant bridge. Each backward ratio tends to $r = \alpha^{-1}$; eventual geometric domination supplies a summable majorant; the fixed prefix vanishes. The resulting normalized sum tends to $1/(1 - r)$. No limit/sum exchange is asserted without its domination condition.

CONDITIONAL — L1.1, L1.9 and P2. Exact external applicability remains EXT-001/002. The paper's latest capacity minimization and Green-function argument are substantially more explicit than in the old supplement. This resolves the earlier compressed local exposition, but does not by itself verify an external theorem's full scope.

MINOR — T1.1 and T1.3. The finite quotient has finite mean absorption: a consecutive outward path has positive probability from every transient state; finitely many states give a uniform positive chance of absorption in a bounded block. Geometric survival bounds justify the first-step mean equations. Empty products equal one. These elementary checks are audit clarifications, not extra source proof text.

MINOR — T2.4. The geometric-tail comparison holds for sufficiently large N. The fixed initial prefix is absorbed into a constant times the eventually increasing positive final term. Proposition 2 already spells out its analogous finite-prefix argument.

Resolved local clarification — L1.4/L1.7. The latest source clips the competing function before expanding the energy and explicitly derives geometric visit counts. The older suggestion to add these explanations should no longer be reported as pending.

NOTE — symmetry and numerics. The latest active Section VI uses plain grouping terminology and omits the old displayed renewal formula. The formula survives only in comments and is not converted into an active

current claim. Empirical accuracy statements do not support any proof step.

Scope discipline. A completed parent step, rather than its private descendants, supplies later unit dependencies. The local transport construction and dimension cases have explicit closure. No circularity or use of the unproved pointwise inequality $Q_D \le H_D$ was found. Every limit, denominator and energy bound is interpreted in its declared domain.

10 Step ledger

The assertion and checked justification for every row appear in the rendering. The current goal is the enclosing unit contract, or a nested parent claim for its children. The contracts supply D, N, p and local domains; definitions and case assumptions remain local. Obligation rows are not successful proof steps.

Step	Goal	Legal dependencies / justification location	Status
G.1	G	contract; checked text at G.1	OK
G.2	G	G.1; checked text at G.2	OK
G.3	G	G.1, G.2; checked text at G.3	OK
G.3.1	G.3	G.1, binomial theorem; checked text at G.3.1	OK
G.3.2	G.3	G.3.1, G.2; checked text at G.3.2	OK
G.4	G	G.1, G.2; checked text at G.4	OK
G.5	G	G.3, G.4; checked text at G.5	OK
G.6	G	G.2, G.3, G.4, G.5; checked text at G.6	OK
T1.1	T1	G; checked text at T1.1	MINOR
T1.2	T1	T1.1, G, G; checked text at T1.2	OK
T1.3	T1	T1.2; checked text at T1.3	MINOR
T1.4	T1	T1.1, T1.2, T1.3; checked text at T1.4	OK
T1.5	T1	T1.4; checked text at T1.5	OK
T1.6	T1	T1.1, T1.5; checked text at T1.6	OK
S.1	S	G, T1; checked text at S.1	OK
S.2	S	G, T1; checked text at S.2	OK
S.3	S	S.2; checked text at S.3	OK
P1.1	P1	contract; checked text at P1.1	OK
P1.2	P1	P1.1, G, G; checked text at P1.2	OK
P1.3	P1	P1.2, G, G; checked text at P1.3	OK
P1.4	P1	P1.1; checked text at P1.4	OK
P1.5	P1	P1.1, P1.3, P1.4; checked text at P1.5	OK
T2.1	T2	T1; checked text at T2.1	OK
T2.2	T2	G, T2.1; checked text at T2.2	OK
T2.3	T2	T2.1, T2.2; checked text at T2.3	OK
T2.4	T2	T1, T2.3; checked text at T2.4	MINOR
T2.5	T2	T2.2, T2.4; checked text at T2.5	OK
R3.1	R3	T2; checked text at R3.1	OK
R3.2	R3	T2, R3.1; checked text at R3.2	OK
R3.3	R3	T1, T2, R3.1, R3.2; checked text at R3.3	OK
R3.4	R3	T2, R3.3; checked text at R3.4	OK
R3.5	R3	R3.4; checked text at R3.5	OK
L1.1	L1	G, P1; checked text at L1.1	CONDITIONAL
L1.O1	L1	source citation; checked text at L1.O1	CONDITIONAL
L1.2	L1	L1.1, P1; checked text at L1.2	CONDITIONAL
L1.3	L1	L1.1, L1.2; checked text at L1.3	CONDITIONAL
L1.4	L1	L1.2, L1.3; checked text at L1.4	CONDITIONAL
L1.5	L1	L1.3, L1.4; checked text at L1.5	CONDITIONAL
L1.6	L1	L1.1, P1; checked text at L1.6	CONDITIONAL
L1.7	L1	L1.1, L1.5, L1.6; checked text at L1.7	CONDITIONAL
L1.8	L1	L1.6, L1.7; checked text at L1.8	CONDITIONAL
L1.9	L1	L1.8, L1.1; checked text at L1.9	CONDITIONAL
L1.O2	L1	source citation; checked text at L1.O2	CONDITIONAL
L1.10	L1	L1.8, L1.9; checked text at L1.10	CONDITIONAL
T3.1	T3	L1; checked text at T3.1	OK
T3.2	T3	T3.1, P1; checked text at T3.2	OK
T3.3	T3	P1, T2, T2; checked text at T3.3	OK
T3.4	T3	L1, T3.2, T3.3; checked text at T3.4	CONDITIONAL
T3.5	T3	T3.4; checked text at T3.5	CONDITIONAL
L2.1	L2	contract; checked text at L2.1	OK
L2.2	L2	L2.1; checked text at L2.2	OK
L2.3	L2	L2.2; checked text at L2.3	OK
L2.4	L2	L2.1; checked text at L2.4	OK

Step	Goal	Legal dependencies / justification location	Status
L2.5	L2	L2.2, L2.4; checked text at L2.5	OK
L2.6	L2	L2.4; checked text at L2.6	OK
L2.7	L2	L2.3, L2.6; checked text at L2.7	OK
L2.8	L2	L2.4, L2.5, L2.7; checked text at L2.8	OK
L2.9	L2	L2.8; checked text at L2.9	OK
P2.1	P2	network convention; checked text at P2.1	CONDITIONAL
P2.2	P2	L2; checked text at P2.2	CONDITIONAL
P2.2.1	P2.2	P2.2, P1; checked text at P2.2.1	OK
P2.2.2	P2.2	P2.2, L2; checked text at P2.2.2	OK
P2.2.3	P2.2	P2.2.1, P2.2.2; checked text at P2.2.3	OK
P2.2.4	P2.2	P2.2.1, P2.2.2, P2.2.3; checked text at P2.2.4	CONDITIONAL
P2.2.5	P2.2	P2.2; checked text at P2.2.5	OK
P2.2.6	P2.2	P2.2.4, P2.2.5; checked text at P2.2.6	CONDITIONAL
P2.3	P2	P2.1, P2.2; checked text at P2.3	CONDITIONAL
T4.1	T4	T3; checked text at T4.1	CONDITIONAL
T4.2	T4	L1, P2; checked text at T4.2	CONDITIONAL
T4.3	T4	T4.1, T4.2, L1; checked text at T4.3	CONDITIONAL
N.1	N	G; checked text at N.1	OK
N.2	N	N.1; checked text at N.2	CONDITIONAL
N.O3	N	source citation; checked text at N.O3	CONDITIONAL
N.3	N	N.2; checked text at N.3	CONDITIONAL
N.4	N	S; checked text at N.4	CONDITIONAL
N.O4	N	source citation; checked text at N.O4	CONDITIONAL

11 Exact-arithmetic diagnostics

The included `check_exact.py` was rerun for this update. All 173 cases passed using Python's exact `fractions.Fraction` arithmetic: 21 one-dimensional flux cases; 12 product-transport cases ($1 \le n \le 3$, $0 \le k \le 3$); 12 pyramid-flow cases ($2 \le D \le 4$, $1 \le N \le 4$); and 128 quotient recurrence cases ($1 \le D \le 4$, $1 \le N \le 8$, $p \in \{1/5, 1/2, 9/10, 1\}$). The flux check covers $0 \le k \le 20$.

The checks exercise endpoint divergence, coordinate-product weights, conservation, factors of two, lateral/forward energy, and the closed formula against the recurrence. They do not prove universally quantified claims, validate the paper's plotted dataset, or close external citations.

12 Minimal clarification plan

Obtain/check EXT-001 and EXT-002 before upgrading the strict true-chain audit verdict. The normalized-sum limit in Remark 3, stray accent command, and unbiased-endpoint wording have been corrected. Retain the finite-mean, empty-product and sufficiently-large-N clarifications when preparing a fully explicit exposition. Check EXT-003/004 separately if their ancillary comparisons require a closed audit. The supplement is mapped to the authorized corrected paper source.

Part III

Reverse dependency audit

13 Audit boundary

The primary conclusion is T4: $H_D(N) = \Theta(\alpha^N/N^{D-1})$ for fixed integer $D \geq 1$ and $0 < p < 1$. Separate roots are the quotient formula, quotient order, transport lemma, Remark 3 and ancillary comparisons. The exact source, hypotheses and external boundary are frozen in Part I. Only supplied routes are followed; no alternative finite-state proof is inserted. The leading-constant completion is now part of the corrected source and is audited explicitly.

14 Obligation graph

Each row is one primary AND route: every listed dependency is required. Cases $D = 1$ and $D \geq 2$ are exhaustive cases whose proofs are both required, not alternative routes that could hide a failed case. No diagnostic or unprovided rescue route is counted as support.

Claim	Primary supplied route	AND dependencies
C0	T4: combine upper and lower bounds	C1, C2, O0
C1	T3: hitting-time lower inequality and shorted shells	C3, C4, O1
C2	T4: hit a face, then apply resistance upper bound	C3, C5, O2
C3	L1: harmonic capacity and killed Green identities	O3, O4, O5, O6
C4	P1/T3: shell cuts in series; retain final term	G, O7, O8
C5	P2: construct unit flow; apply Thomson and sum bound	C6, O6, O9
C6	P2: one-dimensional path and higher-dimensional pyramid	C7, O10, O11
C7	L2: coordinate-by-coordinate redistribution	O12, O13, O14
Q0	T1: unroll differences and telescope	G, O15, O16
Q1	T2: positive finite series and ratio comparison	Q0, O17, O18
Q2	R3: proved sharper equivalent	Q1, O19
Q3	N.2: exact symmetry grouping	O20
Q4	N.4: predecessor parameter comparison	Q0, O21

15 Critical findings

O19 is now discharged by the added Remark 3 argument. It is not reachable from the main root C0. O3 and O6 are external-verification leaves on the true-chain route. The local constructive obligations O10–O14 close independently of these citation checks. No cycle, hidden use of the quotient as the true dynamics, or unjustified inference from numerical accuracy was identified.

16 Dependency table

ID	Claim / obligation	Needed by	Route	Classification	Support	Status
C0	Main true-chain order	root	T4	root	C1,C2,O0	open

ID	Claim / obligation	Needed by	Route	Classification	Support	Status
C1	$H_D \geq 2^{-(D-1)}\alpha^N/N^{D-1}$	C0	T3	major claim	C3,C4,O1	open
C2	$H_D \leq MC'_{D,p}\alpha^N/N^{D-1}$	C0	T4	major claim	C3,C5,O2	open
C3	Hitting-time identity and two resistance inequalities	C1,C2	L1	major claim	O3–O6	open
C4	Shorted-shell resistance lower bound	C1	P1,T3	major claim	G,O7,O8	discharged locally
C5	Face resistance upper bound	C2	P2	major claim	C6,O6,O9	open
C6	Feasible path/pyramid flow and energy	C5	P2	major claim	C7,O10,O11	discharged locally
C7	Nested-cube transport bound	C6	L2	major claim	O12–O14	discharged
G	Shell counts and proposal averages	C4,Q0	G	proved earlier	G.1–G.6	discharged
Q0	Exact finite quotient sum	root,Q1,Q4	T1	root	G,O15,O16	discharged
Q1	Quotient asymptotic order	root,Q2	T2	root	Q0,O17,O18	discharged
Q2	Sharper quotient equivalent	root	R3	root	Q1,O19	discharged
Q3	Symmetry grouping preserves dynamics	root	N.2	root	O20	open
Q4	Predecessor formula correspondence	root	N.4	root	Q0,O21	open
O0	Fixed parameters; constants finite and independent of depth	C0	contracts	side condition	L1.1,P2.2.5	discharged locally
O1	$\mu(0) = 1$; $(2N-1)^{D-1} \leq (2N)^{D-1}$	C1	definitions	algebraic/logical step	T3.4–T3.5	discharged
O2	$F_N^\circ \subset A_N$ nonempty finite; $\tau_N \leq \tau_{F_N^\circ}$	C2	target comparison	definition	V-E; T4.2	discharged
O3	Obtain/check exact recurrence and fixed-state hitting-time background	C3	EXT-001	named theorem	L1.O1	open
O4	Bounded harmonic potential attains capacity	C3	energy expansion	proved earlier	L1.2–L1.5; recurrence via O3	conditional on O3
O5	Killed Green reversibility, occupation sum, geometric visits	C3	Green argument	proved earlier	L1.6–L1.8; recurrence via O3	conditional on O3
O6	Obtain/check infinite-network duality and unit-flow inequality	C3,C5	EXT-002	named theorem	L1.O2	open
O7	Outer shell separates exterior; finite shell shorting reduces resistance	C4	shorting route	named theorem	P1,T3.1–T3.2; finite shorting calculation	discharged locally
O8	Cut conductance; series sum and its final term	C4	series network	algebraic/logical step	P1,T3.3	discharged
O9	$\sum_{j=1}^{N} \alpha^j/j^{D-1} \leq K\alpha^N/N^{D-1}$	C5	eventual ratio	proved earlier	P2.2.5	discharged
O10	Path/forward/lateral conductances and flow energies	C6	explicit construction	definition	P2.1,P2.2.1–P2.2.2	discharged
O11	Current conservation; disjoint energies; dimension and depth cases	C6	construction closure	side condition	P2.2.3–P2.3	discharged
O12	Cumulative one-dimensional flux and energy	C7	cumulative sum	proved earlier	L2.1–L2.3	discharged
O13	Coordinate-line weights and stage divergence	C7	product laws	proved earlier	L2.4–L2.7	discharged
O14	Different coordinate directions; telescoping divergence	C7	stage sum	algebraic/logical step	L2.8	discharged
O15	Finite absorption mean; origin boundary equation	Q0	finite chain	side condition	T1.1; forward clarification	discharged
O16	Recurrence unrolling, product cancellation, final telescoping	Q0	algebra	algebraic/logical step	T1.2–T1.6	discharged
O17	$1 \leq F_\ell \uparrow F_\infty < \infty$	Q1	positive series	proved earlier	T2.1–T2.2	discharged
O18	Ratio below one eventually; finite prefix absorbed	Q1	geometric bound	proved earlier	T2.3–T2.5	discharged
O19	$Q_D(N)/b_{N-1} \to \alpha/(\alpha-1)$	Q2	R3.1–R3.4	proved earlier	Geometric domination and limits	discharged
O20	Obtain/check exact aggregation theorem and applicability	Q3	EXT-003	named theorem	N.O3	open
O21	Obtain/check prior formula and acceptance convention	Q4	EXT-004	unavailable support	N.O4	open

17 Verdict

Main true-chain result: INDETERMINATE FROM THE PROVIDED MATERIAL under the frozen external-source boundary, at O3 and O6. Its local construction is supported, while the invoked external statements remain to be checked. This is not a finding that the theorem is false.

Exact quotient formula, quotient order, sharper equivalent and transport lemma: FOLLOWS, with the minor finite-chain and eventual-threshold clarifications in the forward audit.

Remark 3: FOLLOWS, with O19 discharged by R3.1–R3.4. The unavailable ancillary external comparisons remain indeterminate at O20/O21.

A Repository use and provenance

The package contains this PDF and its LaTeX, the corrected manuscript source snapshot, the source hash, the structured step ledger, the exact-rational script and results, and a small renderer. Run `python3 build.py`, then `pdflatex` twice on `Metropolis_Lamport_Proof_Supplement_Updated.tex`. Run `python3 check_exact.py` for the finite diagnostic suite. Compiling the supplement does not require the paper's figures or bibliography.

The earlier October manuscript source hash began `a41bccc1eb04886d`; it is not the source identity of this update. The source hash above identifies this corrected revision. The earlier uploaded PDF predates these authorized edits and is not the source of this supplement.

1. H. Ramachandran and B. Krishnamachari, *Metropolis Random Walks Escape Deep Basins Faster in Higher Dimensions*, corrected LaTeX manuscript, 10 November 2026. Primary proof source.

2. WWresearch, *Lamport Proof*, v0.2.0. https://github.com/WWresearch/lamport-proof/tree/v0.2.0. Conversion, forward audit and reverse dependency audit. The workflows are LLM instructions, not mathematical validation software.

3. L. Lamport, *How to Write a 21st Century Proof*, 2011, revised 2012. https://lamport.azurewebsites.net/pubs/proof.pdf. Methodological reference.

4. P. G. Doyle and J. L. Snell, *Random Walks and Electric Networks*, 1984; online version 5 July 2006. Sec. 1.3.5 (finite-network Thomson principle), Sec. 2.2.2 (shorting). These passages were checked in the earlier audit; this update does not claim a newly verified infinite-network convention. https://math.dartmouth.edu/~doyle/docs/walkspdf/walks.pdf.

5. J. G. Kemeny and J. L. Snell, *Finite Markov Chains*, 1976, Sec. 6.3, Thm. 6.3.2. Exact external-text verification remains EXT-003.